

Battery Safety Monitoring System

Microprocessors and Embedded Systems Project Report

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1 Project Overview

We developed a battery safety monitoring system using the STM32 Nucleo-L439ZI board. The system continuously monitors pressure, temperature, voltage, and smoke to detect precursors of battery failure or hazardous conditions. One of the STM boards is used to fetch the raw ADC values from the sensor and is displayed using an LCD display. These values are transmitted to another STM32 board using UART, where all the processing is being done to demonstrate the precursors. Thus, the second LCD display shows the real values, which are mounted in front of the model.

2 Components Used

- STM32 Nucleo-L439ZI (x2)
- I2C 16x2 LCD Display (x2)
- Pressure Sensor (Piezoelectric Sensor)
- Temperature Sensor (LM35)
- Smoke Sensor (MH MQ-2)
- Buzzer

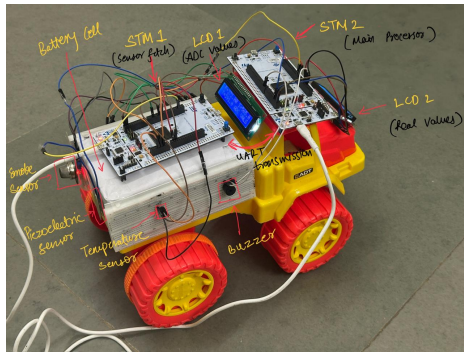


Figure 1: Battery Safety System

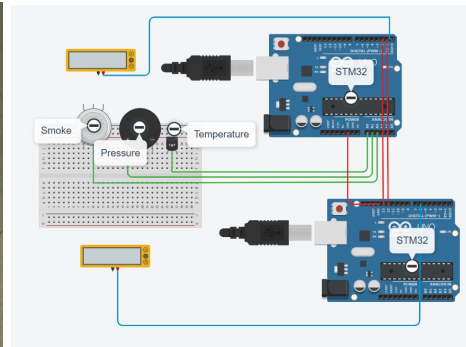


Figure 2: Schematic

3 Method

All the sensor values are read via an ADC using different channels. A timer triggers LCD updates. Then, the mapping is done from the raw ADC values to the realistic values using a linear function. There are 3 different states- Safe, Cooldown and Evacuation. In the safe condition, the sensors on the battery detect nominal values, while in cooldown condition, the battery needs to cool down or needs to be checked. Lastly, the evacuation condition shows that the battery is going to explode and thus, the vehicle should be evacuated. The conditions are as follows:

- **Cooldown:** Temperature above 70°C and Voltage below 3.3 V
- **Evacuation:** There are different conditions for evacuation:
 - Smoke is detected above 700 ppm
 - Temperature exceeds 95°C
 - Pressure exceeds 3 atm for a specific amount of time.

4 Results

We tested our model in different conditions and observed that it responded appropriately in all the conditions and updated the LCD. [Final Demonstration Video](#)

5 References

- [Research Presentation](#)
- [I2C](#)
- [ADC](#)
- [UART](#)